CUSTOMER CHURN ANALYSIS

Week 8 Capstone Project | The Developers Arena Internship

Business problem

Customer churn represents customers who stop using a company's service. This project examines the supplied customer dataset to measure churn and explore customer characteristics associated with leaving. The objective is to turn the data into practical retention actions.

Dataset and quality

The supplied dataset contains **500 rows** and **9 columns**. The analysis checks missing values, duplicate records, data types, and descriptive statistics. The raw file is retained separately from the cleaned working copy.

Customers	Churned	Retained	Churn rate
500	53	447	10.6%

Analysis approach

The workflow combines descriptive statistics, customer-segment comparisons, visual exploratory data analysis, and (where the dataset fields support it) a classification model. Segment churn rates are calculated as the share of customers in each segment with a churn outcome. Model results should be interpreted with precision, recall, F1-score and the confusion matrix—not accuracy alone.

Key observations

• Observed churn rate in this sample: **10.6%** (53 of 500 customers with a valid churn label).
• Highest observed contract-segment churn rate: **Month-to-month** (20.6%).
• Highest observed payment-method churn rate: **Credit Card** (13.5%).
• Average customer tenure: **36.5** (dataset units).
• Average monthly charge: **113.64** (dataset currency/units).

Recommended business actions

Prioritise outreach to customer segments with higher observed churn; test retention offers rather than applying blanket discounts; review contract and payment experiences; monitor early-tenure customers; and track churn rate over time. Treat these as hypotheses to validate through controlled campaigns and additional data.

Limitations

This is a sample-based analysis. Associations do not prove causation, and the dataset may not represent the full customer population. Confirm field definitions, time period, currency, and business context with the data owner before operational use.